Xiaosheng Huang

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Education

PhD, Physics: University of California, Berkeley	December, 2004
MA, Physics, Kent State University	May, 1995
BS, Physics (Summa Cum Laude), Kent State University	May, 1995

Professional Appointments

Professor: Department of Physics & Astronomy, University of San Francisco	2024 –
Associate Professor: Department of Physics & Astronomy, University of San Francisco	2018 – 2024
Assistant Professor: Department of Physics & Astronomy, University of San Francisco	2012 – 2018
Lecturer: Physics Department, University of California, Berkeley	2007 – 2012
Postdoctoral Fellow: UC Berkeley/Lawrence Berkeley National Laboratory	2005 – 2012
Graduate Student Researcher: Physics Department, UC Berkeley	2000 – 2004
Graduate Student Researcher: Astronomy Department, UC Berkeley	1996 – 1999[1]
Graduate Student Instructor: Physics Department, UC Berkeley	1995, 1999
Teaching Assistant: Physics Department, Kent State University	1993 – 1995
Lab Assistant: Liquid Crystal Institute, Kent State University	1992 – 1993

Press Releases & Publicity

NPR Interview	2025
Lawrence Berkeley National Laboratory Press Release	2024
USF News Article on Roman Space Telescope Program	2023
Lawrence Berkeley National Laboratory Computing Sciences Highlight	2023
European Southern Observatory Press Release	2023
American Astronomical Society Nova Highlight	2023
Press Conference, American Astronomical Society Annual Meeting	2021
American Astronomical Society Nova Highlight	2021
National Science Foundation's NOIRLab Press Release	2021
Lawrence Berkeley National Laboratory Press Release	2021
Lawrence Berkeley National Laboratory Press Release	2020
USF Magazine Article on Faith	2020
USF News Article on Hubble Space Telescope Program	2019

Awards & Grants

Hubble Space Telescope Program *GO-18085,* with NASA Grant ($190K projected)	2025
Hubble Space Telescope Program *AR-18151,* with NASA Grant ($60K projected)	2025
Dean's Scholar Award, College of Arts & Sciences, USF	2024
NASA Roman Space Telescope Mission Research and Support Program ($64K)	2023
Outstanding Mentors Award, Lawrence Berkeley National Laboratory	2021
Hubble Space Telescope Program *GO-15867*, with NASA Grant ($189K)	2019
Visiting Faculty Program, Department of Energy (with summer stipends)	2017, 2018, 2019, 2025
Faculty Development Fund, USF (multiple awards each year)	2012 –
Graduate Student Travel Grant, University of California, Berkeley	2004
University Fellowship, University of California, Berkeley	1995

[1] My initial graduate research, with T. Broadhurst, focused on gravitational lensing. After his departure from the U.S., I completed my PhD in condensed matter physics (with A. Zettl), and later returned to cosmology as a postdoctoral researcher at Lawrence Berkeley National Laboratory, working on dark energy.

Research

Research Highlights

Among the biggest questions in cosmology are the nature of dark energy and dark matter and the expansion rate of the universe. To address these questions, I lead the **multi-institutional** *DESI Strong Lens Foundry* within the Dark Energy Spectroscopic Experiment (DESI) and collaborate closely with the *Supernova Cosmology Project* led by Prof. Saul Perlmutter, both based at Lawrence Berkeley National Laboratory (LBNL). Using DESI imaging data, my team has discovered over 5000 strong lens candidates, featured in multiple press releases and a national press conference. We also found 1346 lens candidates in the UNIONS imaging data. In DESI spectroscopy, using "classical" methods, we found an additional 6000 candidates, reaching a grand total of 12,000. We developed a GPU-accelerated Bayesian lens modeling framework that underpins our current and forthcoming Hubble Space Telescope (HST) and JWST programs. I am PI on three HST programs with NASA support ($189K in 2019; $250K projected in 2025) and supervise two additional programs for which I serve as Co-I. I am also Co-PI/Co-I on major observing programs at the Gemini, Keck, and Very Large Telescopes, as well as on space-based programs with the Nancy Grace Roman Space Telescope ($64K NASA grant) and JWST. We were recently selected by the competitive NERSC Science Acceleration Program (NESAP), with early access to the next-generation supercomputer *Doudna*, with 12,000 GPU hours awarded and NERSC engineers dedicated to helping adapt and scale `GIGA-Lens` to the new architecture. Finally, we are preparing a targeted search for lensed supernovae to measure H_0, with a paper demonstrating accurate and precise modeling of lensed point sources using `GIGA-Lens` soon to be submitted.

Selected publications and lens candidate catalogues are available at our project website.

Principal Investigator, Successful Telescope Proposals

- PI, HST, 2025 (Cycle 33): Confirming Spectroscopic Strong Lenses Found in New Discovery Windows from DESI DR1
- PI, HST, 2025 (Cycle 33): Finding Low Halo-Mass Strong Lenses in the HST Archive
- Co-PI, Keck Observatory, 2022–2025: Redshifts and Modeling for Low-Mass Halo Detection
- Co-PI, DESI Secondary Target Program, 2021: Spectroscopic Confirmation of Lens Candidates
- PI, HST, 2019 (Cycle 27): Confirming Strong Galaxy Lenses in DESI Legacy Surveys

Co-Investigator, Successful Telescope Proposals

- HST:
  - 2024, Confirming Spectroscopic Strong Lens Candidates from the DESI One-Percent Survey (PI: A. Bolton)
  - 2024, Automated Pipeline for Modeling Strong Lenses Observed by HST (PI: C. Storfer)
  - 2020, LensWatch: Time-Delay Measurement of a Multiply-Imaged Supernova (PI: J. Pierel)
  - 2016–2017, SUbaru Supernovae with Hubble Infrared (SUSHI) (PI: N. Suzuki)
  - 2014–2015, See Change: Testing Time-Varying Dark Energy with $z > 1$ Supernovae (PI: S. Perlmutter)
- Alma APEX: 2024, ASADOS (PI: K. Harrington)
- Gemini: 2024, Gravitational Lenses in UNIONS and Euclid (GLUE): Spectroscopic Confirmation of Strongly Lensed Galaxies (PI: C. Storfer)
- Nancy Grace Roman Space Telescope: 2023 (NASA ROSES-D.14), Enabling Lensed SN Cosmology (PI: J. Pierel)

- Joint HST + JWST Program (2023): Pioneering Precision: Advancing Cosmology with the First Statistical Sample of Lensed Supernovae
- JWST (2022, Cycle 1): Imaging and Spectroscopy of Three Highly Magnified Images of a Supernova at $z = 1.5$ (PI: P. Kelly)
- VLT / MUSE: DESJ0603: A uniquely spherical, multi-source lensing cluster perfect for cosmology (PI: T. Barone, 2024)
- VLT / MUSE: Characterization of galaxy-galaxy gravitational lens systems with MUSE (PIs: A. Cikota, F. Bian 2021 - 2023)
- VLT / Xshooter: Strong lensing and mass assembly (2021–2022, PI E. Jullo)
- NSF NOIRLab DECam / Blanco: Imaging and transient surveys (2019–2023; PIs D. Schlegel, A. Palmese, I. Andreoni)
- Keck Observatory: Multiple programs, 2010–2018

Principal Investigator, Successful Supercomputer Proposals

- PI, 2026–2027, NESAP selection
- Co-PI, multiple proposals 2020–2026, NERSC "regular" allocations (ERCAP)

Teaching

Teaching Highlights

I developed and continue to update the ML/AI course series PHYS 301–303 and founded two minor programs: *Engineering Physics* (2016–) and *Applied AI* (launching 2025). Courses have been adapted/cross-listed at the master's level across Computer Science and Economics.

Courses, University of San Francisco (2012–)

USF	PHYS 120/121 (Intro. to Astronomy); 301–303 (AI series); 343 (Cosmology); 350 (Colloquium); 371 (Mathematical Methods for Physicists); ENGR 302 (Computation).

Courses, University of California, Berkeley (2007–2012, Lecturer)

Berkeley	Lower Div: Honors 7B (Gen. Phys. II), 7C (Gen. Phys. III), 8A (Pre-med Gen. Phys.); Upper Div: 112 (Stat Mech), 137 (Quantum Mechanics)

Development of New Courses

- **PHYS 301: Introduction to Scientific Computation** (2013; major revision 2015). Python-based modernization; added lab; expanded to Monte Carlo, DFT, PCA, SVM; M.S. cross-listing in CS (2022).
- **PHYS 302: Scientific Computation & Machine Learning** (2016). Added decorators, regression, ANNs, CNN, ODE systems, symbolic computation; M.S. cross-listings in CS/Econ (2020, 2023; RL/ResNets).
- **PHYS 303: Deep & Bayesian Learning** (2023). Deep architectures (e.g., Transformer) and Bayesian inference (HMC, VI, latent variables, high-dimensional optimization); M.S. adaptation for CS (2025).
- **ENGR 302: Scientific Computation for Engineers** (2022). Two-unit engineering applications course.

- **PHYS 343: Astrophysics (Cosmology)** (2013). New cosmology course under existing number.

Service

Service Highlights

Mentored **50+** undergraduate researchers at USF, UC Berkeley and other universities, leading to peer-reviewed publications and national presentations; received LBNL's Outstanding Mentors Award (2021). Lead the multi-institutional *DESI Strong Lens Foundry* collaboration.

Service to University/College

- **Curriculum:** Engineering curriculum development (2016–); joint appointment (2020–2024).
- **Committees:** Dean's Award Review (2025); Distinguished Research (member 2014; Chair 2015–2016); Honorary Degree (2014–2018); Learning Technologies (2015–2018); Learning Analytics (2017–2018).
- **Outreach:** Destination USF Mock Class (2016).

Service to Department/Students

- **Mentoring:** Research Supervision since 2013; multiple ApJ/ApJS publications with students (undergraduate and graduate) co-authors/lead authors; CS M.S. capstone mentor (2018, 2019, 2022).
- **Student Presentations:** AAS (incl. 2020); USF CARD (2014, 2015, 2018, 2019, 2023, 2024); USF events: Dean's Circle (2022), President's Inauguration (2014), Innovation Symposium (2013).
- **Fellowships:** Guided successful USF Summer Research Fellowship proposal (2015).
- **Programs:** Engineering Physics Minor (2017–; 30+ recipients); Applied AI Minor (launching 2025).
- **Hiring:** Faculty search committees (tenure lines): 2016, 2018.

Professional Activities & Service

- Lead, *DESI Strong Lens Foundry* (2019–).
- LBNL student mentorship (2013–; DOE SULI); LBNL *Outstanding Mentors Award* (2021).
- PhD Thesis Committee, IfA (U. Hawai'i, 2023–); Senior Thesis Committee, Yale (2024–2025).
- HST proposal reviewer (MidCycle 27; Cycle 29).
- Journal referee: *Nature Astronomy*, *Space Science Review*, *ApJS*, *AJ*, *MNRAS*.
- Invited talks: UC Berkeley Astronomy Colloquium (2025); Santa Clara Univ. Physics Colloquium (2025); UC Davis Cosmology Seminar (2025); University of Rochester Colloquium (2024); USF Colloquium (2023); LBNL Machine Learning Seminar (2022); KIPAC/Stanford Cosmology Seminar (2022); NERSC Data Seminar (2022); LBNL Physics Division Seminar (2021); Florida State University Astrophysics Seminar (2020); USF Colloquium (2017); San Jose State Univ. Colloquium (2016); Wonderfest Public Lecture (2016); Sonoma State Univ. "What Physicists Do" Series (2015).
- Conferences Presentations: Science with Strong Lensing, ALMA and Next-Generation Radio Interferometry (NRAO, Charlottesville, 2025); Strong Gravitational Lensing Science Conference (CfA/Harvard-Smithsonian, 2025); Strong Lensing with LSST (Oxford, 2024); DESI Collaboration Meeting Plenary (2024); American Astronomical Society Annual Meetings (2014–2021); Nearby Supernova Factory Collaboration Meeting (Clermont-Ferrand, 2017).

Peer-reviewed Publications[2]

1. *E. Lin†, I. Bertolla, A. Cikota, **X. Huang**, et al. 2025, submitted to ApJS, arXiv:2509.18078
DESI Strong Lens Foundry IV: Spectroscopic Confirmation of DESI Lens Candidates with VLT/MUSE
2. *S. Agarwal†, **X. Huang**, et al. 2025, submitted to ApJ, arXiv:2509.18086
DESI Strong Lens Foundry III: Keck Spectroscopy for Strong Lenses Discovered Using Residual Neural Networks
3. **X. Huang**, Jose Inchausti, C. Storfer, S. Tabares-Tarquinio, et al. 2025, submitted to ApJ, arXiv:2509.18089
DESI Strong Lens Foundry II: DESI Spectroscopy for Strong Lens Candidates
4. *Y. Hsu†, **X. Huang**, et al. 2025, submitted to ApJ, arXiv:2509.16033
A New Way to Discover Strong Gravitational Lenses: Pair-wise Spectroscopic Search from DESI DR1
5. *Jose Inchausti, C. Storfer, **X. Huang**, et al. 2025, submitted to ApJ, arXiv:2508.20087
Strong Lens Discoveries in DESI Legacy Imaging Surveys DR10 with Two Deep Learning Architectures
6. C. Storfer, E.A. Magnier, **X. Huang**, et al. 2025, submitted to ApJ, arXiv:2505.05032
Gravitational Lenses in UNIONS and Euclid (GLUE) I: A Search for Strong Gravitational Lenses in UNIONS with Subaru, CFHT, and Pan-STARRS Data
7. T. Barone, et al. 2025, AJ accepted, arXiv:2503.08041
The AGEL Survey Data Release 2: A Gravitational Lens Sample for Galaxy Evolution and Cosmology
8. **X. Huang**, S. Baltasar†, N. Ratier-Werbin†, et al. 2025, ApJ accepted, arXiv:2502.03455
DESI Strong Lens Foundry I: HST Observations and Modeling with GIGA-Lens
9. *E. Silver†, R. Wang†, **X. Huang**, A. Bolton, et al., 2025, ApJ, 994, 117
ML-Driven Strong Lens Discoveries: Down to $\theta_E \sim 0.03''$ *and* $M_{\rm halo} < 10^{11} M_\odot$
10. D. Rubin et al. 2025, ApJ, 986 231
Union through UNITY: Cosmology with 2000 SNe Using a Unified Bayesian Framework
11. B. Wedig et al. 2025, ApJ, 986, 42
The Roman View of Strong Gravitational Lenses
12. E. Khalouei et al. 2025, A&A, 698, 266
Detection of unresolved strongly lensed supernovae with the 7-Dimensional Telescope
13. F. Urcelay et al. 2025, A&A, 694, A35
A compact group lens modeled with GIGA-Lens: Enhanced inference for complex systems
14. J.D.R. Pierel et al. 2024a, ApJ, 967, 50
JWST Photometric Time-delay and Magnification Measurements for the Triply Imaged Type Ia "SN H0pe" at z = 1.78
15. J.D.R. Pierel et al. 2024b, ApJL 967 L37
Lensed Type Ia Supernova "Encore" at z = 2: The First Instance of Two Multiply Imaged Supernovae in the Same Host Galaxy
16. W. Sheu, A. Cikota, **X. Huang** et al., 2024, ApJ, 973, 3
The Carousel Lens: A Well-Modeled Strong Lens with Multiple Lensed Sources

[2] * = Papers led by a student who was an undergraduate at least at the start of the project. An underline indicates undergraduate co-authors from USF, and a †, those from other institutions.

17. *W. Sheu[†], **Xiaosheng Huang**, et al., 2024, ApJ, 973, 24
A Targeted Search for Variable Gravitationally Lensed Quasars

18. *C. Storfer, **X. Huang**, et al., 2024, ApJS, 274, 16
New Strong Gravitational Lenses from the DESI Legacy Imaging Surveys Data Release 9

19. *C. Dawes[†], C. Storfer, **X. Huang**, G. Aldering, A. Dey, D. J. Schlegel, 2023, ApJS, 269, 61
Finding Multiply-Lensed and Binary Quasars in the DESI Legacy Imaging Surveys

20. A. Cikota, I. Bertolla, **X. Huang**, et al., 2023, ApJL, 953, L5
A New Einstein Cross Spectroscopically Confirmed with VLT/MUSE and Modeled with GIGA-Lens

21. A. Filipp, Y. Shu, R. Pakmor, S. H. Suyu, and **X. Huang**, 2023, A&A, 677, A113
Simulation-guided Galaxy Evolution Inference: A Case Study with Strong Lensing Galaxies

22. D. Langeroodi, et al., 2023, ApJ, 957, 39
Evolution of the Mass-Metallicity Relation from Redshift $z \approx 8$ to the Local Universe

23. *W. Sheu[†], **X. Huang**, A. Cikota, N. Suzuki, D.J. Schlegel, C. Storfer, ApJ, 952, 10, 2023
Retrospective Search for Strongly Lensed Supernovae in the DESI Legacy Imaging Surveys

24. J.D.R. Pierel, N. Arendse, S. Ertl, **X. Huang**, L. A. Moustakas, S. Schuldt, A. J. Shajib, Y. Shu, S. Birrer, M. Bronikowski, J. Hjorth, S. H. Suyu, et al., 2023, ApJ, 948, 115
LensWatch: I. Resolved HST Observations and Constraints on the Strongly-Lensed Type Ia Supernova 2022qmx ("SN Zwicky")

25. *A. Gu[†], **X. Huang**, et al., ApJ, 935, 17, 2022
GIGA-Lens: Fast Bayesian Inference for Strong Gravitational Lens Modeling

26. B. Hayden, et al., ApJ, 912, 87, 2021
The HST See Change Program: I. Survey Design, Pipeline, and Supernova Discoveries

27. **X. Huang**, C. Storfer, A. Gu, V. Ravi, A. Pilon, et al., ApJ, 909, 27, 2021
Discovering New Strong Gravitational Lenses in the DESI Legacy Imaging Surveys

28. **X. Huang**, M. Domingo, A. Pilon. V. Ravi, C. Storfer, D.J. Schlegel, et al., ApJ, 894, 78, 2020
Finding Strong Gravitational Lenses in the DESI DECam Legacy Survey

29. J. Kim, M.J. Jee, S. Perlmutter, B. Hayden, D. Rubin, **X. Huang**, G. Aldering, J. Ko, ApJ, 887, 76, 2019
Precise Mass Determination of SPT-CL J2106-5844, the Most Massive Cluster at $z > 1$

30. R. Rubin, B. Hayden, X. Huang, ..., Z. Raha, et al., ApJ, 866, 65, 2018
The Discovery of a Gravitationally Lensed Supernova Ia at Redshift 2.22

31. **X. Huang**, G. Aldering, M. Biederman, and B. Herger, ApJ, 850, 84, 2017
On the Time Variation of Dust Extinction and Gas Absorption for Type Ia Supernovae Observed through a Nonuniform Interstellar Medium

32. **X. Huang**, Z. Raha, et al., ApJ, 836, 157, 2017
The Extinction Properties of and Distance to the Highly Reddened Type Ia Supernova SN 2012cu

33. J. Nordin, et al., MNRAS, 440, 2742, 2014
Lensed Type Ia supernovae as probes of cluster mass models

34. D. Rubin, et al., ApJ, 763, 35, 2013
Precision Measurement of The Most Distant Spectroscopically Confirmed Supernova Ia with the Hubble Space Telescope

35. N. Suzuki, et al., ApJ, 746, 85, 2012
The Hubble Space Telescope Cluster Supernova Survey: V. Improving the Dark Energy Constraints Above $z > 1$ and Building an Early-Type-Hosted Supernova Sample

36. K. Barbary, et al., ApJ, 745, 31, 2012
The Hubble Space Telescope Cluster Supernova Survey. VI. The Volumetric Type Ia Supernova Rate

37. K. Barbary, et al., ApJ, 745, 32, 2012
The Hubble Space Telescope Cluster Supernova Survey: II. The Type Ia Supernova Rate in High-Redshift Galaxy Clusters

38. M.J. Jee, et al., ApJ, 737, 59, 2011
Scaling Relations and Overabundance of Massive Clusters at $z \gtrsim 1$ from Weak-lensing Studies with the Hubble Space Telescope

39. **X. Huang**, et al., ApJL, 707, L12, 2009
Hubble Space Telescope Discovery of a z = 3.9 Multiply Imaged Galaxy Behind the Complex Cluster Lens Warps J1415.1+36 at z =1.026

40. K.S. Dawson, et al., AJ, 138, 1271, 2009
An Intensive HST Survey for $z > 1$ Type Ia Supernova by Targeting Galaxy Clusters

41. K. Barbary, K., et al., ApJ, 690, 1358, 2009
Discovery of an Unusual Optical Transient with the Hubble Space Telescope

42. **X. Huang**, W. Mickelson, B.C. Regan, Steve Kim†, A. Zettl, 2006, Solid State Communications, 140, 163
Pressure Dependence of Tc and Hc2 of a Dirty Two-gap Superconductor, Carbon-doped MgB2

43. **X. Huang**, W. Mickelson, B.C. Regan, and A. Zettl, 2005, Solid State Communications, 136, 278
Enhancement of the Upper Critical Field of MgB2 by Carbon-doping

44. W. Han, J. Cumings, **X. Huang**, et al., Chemical Physics Letters, 346, 368, 2001
Synthesis of Aligned BxCyNz Nanotubes by a Substitutionreaction Route

45. T. Broadhurst, **X. Huang**, B. Frye, and R. Ellis, ApJL, 534, L15, 2000
A Spectroscopic Redshift for the Cl 0024+16 Multiple Arc System: Implications for the Central Mass Distribution

Publications in Prep

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46. C. Storfer, M. Tamargo†, **X. Huang**, et al. 2025
DESI Strong Lens Foundry V: Keck Spectroscopy for 15 Strong Lenses Discovered Using Residual Neural Network

47. **X. Huang**, D. Alvarez†, M. Ubeda†, et al. 2025, in prep.
DESI Strong Lens Foundry VI: An HST Sample of Strong Lenses Modeled with GIGA-Lens

48. *S. Baltasar†, N. Ratier-Werbin†, **X. Huang**, et al. 2025, in prep.
Modeling of Multiply Lensed Point Sources

49. F. Urcelay, **X. Huang**, et al. 2025, in prep.
Cosmology with the Cosmic Carousel Lens

50. *J. Karp†, D.J. Schlegel, **X. Huang**, et al. 2025, in prep.
The DESI Single Fiber Lens Search: Four Thousand Spectroscopically Selected Galaxy-Galaxy Gravitational Lens Candidates

51. *N. Ratier-Werbin†, H. Lu†, L. Upton†, **X. Huang**, et al. 2025, in prep.
Lens Modeling with Multiple GPU Nodes

52. C. Storfer, **X. Huang**, et al. 2025, in prep.
DESI Strong Lens Foundry VII: A Catalog of 1000 Spectroscopically Confirmed Strong Gravitational Lenses in DESI Year 3 Data Release

Other Publications

- Pierel, J. et al., Transient Name Server AstroNote 2022-196 Multiple Images of SN 2022qmx ("SN Zwicky") Resolved in HST Observations
- Boone, K. et al., 2016ATel. 9125 SCP16L01: discovery of an unusual transient in MOO-J1142